

## **CURRICULUM VITAE**

### **Dr. Gregory D. Graff**

Department of Agricultural & Resource Economics  
College of Agricultural Sciences  
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### **EDUCATION**

|      |                                                                               |
|------|-------------------------------------------------------------------------------|
| 2002 | Ph.D., Agricultural and Resource Economics, University of California Berkeley |
| 1999 | M.S., Agricultural and Resource Economics, University of California Berkeley  |
| 1995 | M.A., Slavic and East European Studies, Ohio State University                 |
| 1992 | B.A., Biological Sciences, Cornell University                                 |

### **ACADEMIC POSITIONS**

|                |                                                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2018 - present | Professor, Agricultural and Resource Economics, Colorado State University, Fort Collins, Colorado, USA.                                                                                        |
| 2015 - 2016    | Visiting Professor, Chalmers School of Entrepreneurship, Department of Technology Management and Economics, Chalmers University of Technology, Gothenburg, Sweden.                             |
| 2011 - 2018    | Associate Professor, Agricultural and Resource Economics, Colorado State University, Fort Collins, Colorado, USA.                                                                              |
| 2007 - 2011    | Assistant Professor, Agricultural and Resource Economics, Colorado State University, Fort Collins, Colorado, USA.                                                                              |
| 2004 - 2007    | Senior Research Associate and Program Director, Public Intellectual Property Resource for Agriculture (PIPRA), Department of Plant Sciences, University of California, Davis, California, USA. |
| 2004 - 2007    | Visiting Research Fellow, Department of Agricultural and Resource Economics, University of California, Berkeley, California, USA.                                                              |

### **SABBATICALS**

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| 2015 - 2016 | Chalmers School of Entrepreneurship, Department of Technology Management and Economics, Chalmers University of Technology, Gothenburg, Sweden. |
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## OTHER POSITIONS

- 2020 - 2023 Expert Advisor, Office of the Chief Economist, United States Patent and Trademark Office (USPTO)
- 2003 - 2004 Director of Research, Bio Economic Research Associates LLC (bio-era), Cambridge, MA
- 2001 - 2003 Senior Analyst, Office of Technology Transfer, University of California, Office of the President, Oakland, CA

## INSTITUTE/PROGRAM AFFILIATIONS

- 2023- present Faculty Affiliate, Institute for Entrepreneurship, Colorado State University
- 2021- present Faculty Affiliate, Soil Carbon Solutions Center, Colorado State University (<https://www.research.colostate.edu/cip/scsc/>)
- 2016 - present Faculty Affiliate, Food Systems, Colorado State University (<http://foodsystems.colostate.edu/people/faculty-affiliates/>)
- 2016 - present Research Associate, Regional Economic Development Institute (REDI), Colorado State University (<http://redi.colostate.edu/bios-and-contact/>)
- 2009 - present GDPE Faculty, Graduate Degree Program in Ecology (GDPE), Colorado State University, Colorado State University (<https://ecology.colostate.edu/faculty-directory/>)
- 2008 - present InSTePP Fellow, International Science and Technology Practice and Policy (InSTePP), University of Minnesota (<https://instepp.umn.edu/about/people>)

## CURRENT JOB DESCRIPTION

40% Teaching, 35% Research/Creative Activity, 25% Service/Extension [updated]

## CURRENT WORKING PAPERS

1. Aaisha Al Maamari and Gregory Graff, *Adoption and diffusion of drought-tolerant corn varieties in the U.S. Corn Belt*, Colorado State University, 2024.
2. Per Hulthén and Gregory Graff, *The financing lifecycle and exits of Europe's pharma, biotech, and health science startups*, Chalmers University of Technology, Gothenburg, Sweden 2024.
3. Per Hulthén and Gregory Graff, *The financing lifecycle and exits of Europe's cleantech startups*, Chalmers University of Technology, Gothenburg, Sweden 2024.

## PUBLISHED WORKS

### Books

1. Anatole Krattiger, Richard Mahoney, Lita Nelson, Jennifer Thompson, Alan Bennett, Kanikaram Satyanarayana, Gregory Graff, Carlos Fernandez, and Stanley Kowalski, Eds., 2007, *Intellectual Property Management in Health and Agricultural Innovation: A Handbook of Best Practices*, Volumes 1 and 2, Oxford, UK: MIHR and Davis, CA: PIPRA, 1998 pages. (Full open access online at <http://www.iphandbook.org/>)

### Refereed Journal Articles

1. Gregory D. Graff, Charles deGrazia, and Nicholas Rada, "The role of venture-financed startups in innovation for U.S. agriculture" forthcoming in *American Journal of Agricultural Economics*, 2026.
2. Gregory Graff and Jacob Sherkow, "Models of technology transfer for genome-editing technologies," *Annual Review of Genomics and Human Genetics*, Vol. 21 (August 2020), pp. 509-534.
3. Gregory Graff and Philip Pardey, "Inventions and patenting in Africa: Empirical trends from 1970 to 2010" *Journal of World Intellectual Property*, 23(1-2), (March 2020), pp. 40-64.
4. Knut Egelie, Gregory Graff, Sabina Strand, and Berit Johansen, "The Emerging Patent Landscape of CRISPR/Cas Gene Editing Technology: Early views on control and access for follow-on commercial applications" *Nature Biotechnology*, 34(10), (October 2016), pp. 1025-1031.
5. Liaw Batan, Gregory Graff, and Thomas Bradley, "Techno-economic and Monte Carlo probabilistic analysis of microalgae biofuel production system," *Bioresource Technology*, 219, (2016), pp. 45–52.
6. Steven Albers, Annabelle Berklund, and Gregory Graff, "The Rise and Fall of Innovation in Biofuels: the global landscape of biofuel patenting from 1970 to 2013," *Nature Biotechnology*, 34(8), (August 2016), pp. 814-821.
7. David Jefferson, Gregory Graff, Cecilia Chi-Ham and Alan Bennett, "The Emergence of Agbiogenetics," *Nature Biotechnology* Vol. 33, (August 2015) pp. 819–823.
8. Gregory Graff, Gal Hochman, Chubashini Suntharlingam, and David Zilberman, "The Competing Policy Paradigms of Agricultural Biotechnology: Implications and Opportunities for Emerging and Developing Economies," *AgBioForum*, (August 2015).
9. David Zilberman, Gregory Graff, Gal Hochman and Scott Kaplan, "The political economy of biotechnology," *German Journal of Agricultural Economics*, Volume 64 (2015), Number 4, pp. 212-223.
10. Chubashini Suntharalingam, Shanmugavelu Sithambaram, Gregory Graff, and Nor Azlina Saari, "Characterizing Innovation in the Livestock Industry: A Malaysian Case" *International Journal of Tropical Agriculture*, Vol. 33, No. 2, (April-June 2015), pp. 879-885.
11. David Zilberman, Scott Kaplan, Eunice Kim, Gal Hochman, and Gregory Graff, "Continents Divided: Understanding Differences between Europe and North America in Acceptance of GM Crops," *GM Crops and Food*, 4(3), (July-December 2013), pp. 202-208.
12. Gregory Graff, Devon Phillips, Zhen Lei, Sooyoung Oh, Carol Nottenburg, & Philip Pardey, "Not Quite a Myriad of Gene Patents," *Nature Biotechnology* Vol. 31, (May 2013) pp. 404-410.
13. Tania Bubela, Richard Gold, Gregory Graff, Daniel Cahoy, Dianne Nicol, David Castle, "Patent Landscaping for Life Sciences Innovation: Toward Consistent and Transparent Practices" *Nature Biotechnology*, Vol. 31, (March 2013) pp. 202-206.
14. Debra Mathews, Gregory Graff, Krishanu Saha, and David Winickoff, "Access to Stem Cells and Data: Persons, Property Rights, and Scientific Progress," *Science*, Vol. 331. (11 February 2011), pp. 725-727.

15. Gregory Graff, David Zilberman, and Alan Bennett, "The Commercialization of Biotechnology Traits," *Plant Science*, 179, (2010) pp. 635-644.
16. Gregory Graff, David Zilberman, and Alan Bennett, "The Contraction of Agbiotech Product Quality Innovation," *Nature Biotechnology*, Vol. 27, No. 8 (August 2009), pp. 702-704.
17. Gregory Graff, Gal Hochman, and David Zilberman, "The Political Economy of Agricultural Biotechnology Policies," *AgBioForum*, Vol. 12, No. 1 (2009), pp. 34-46.
18. David Winickoff, Kris Saha, and Gregory Graff, "Opening Stem Cell Research: A Policy Proposal for the Management of Data, Intellectual Property, and Ethics," *Yale Journal of Health Law, Policy and Ethics*, Vol. IX, No. 1 (Winter 2009), pp. 52-127.
19. Karl Bergman and Gregory Graff, "The Global Stem Cell Patent Landscape: Implications for Efficient Technology Transfer and Commercial Development" *Nature Biotechnology*, Vol. 25, No. 4 (April 2007), pp. 419-425.
20. Gregory Graff, David Roland-Holst, and David Zilberman, "Biotechnology and Poverty Reduction in Low-Income Countries," *World Development*, Vol. 34, No. 8 (2006): 1430-1445.
21. David Zilberman and Gregory Graff "Intellectual Property Rights, Innovation, and the Evolution of Biotechnology in Developing Countries," *Quarterly Journal of International Agriculture*, Vol. 44, No. 3 (2005), pp. 247-266.
22. David Zilberman, Holly Ameden, Gregory Graff, and Matin Qaim, "Agricultural Biotechnology: Productivity, Biodiversity, and Intellectual Property Rights," *Journal of Agricultural & Food Industrial Organization*, Vol. 2, No. 2 (2004), Article 3.
23. Gregory Graff, Brian Wright, Alan Bennett, and David Zilberman, "Access to Intellectual Property: an Obstacle to Developing Transgenic Horticultural Crops," *California Agriculture*, (April-June 2004), pp. 121-126.
24. Gregory Graff, "Observing Technological Trajectories in Patent Data: Empirical Methods to Study the Emergence and Growth of New Technologies," *American Journal of Agricultural Economics*, Vol. 85, No. 5 (2003), pp. 1266-1274.
25. Deborah Delmer, Carol Nottenburg, Gregory Graff, and Alan Bennett, "Intellectual Property Resources for International Development in Agriculture," *Plant Physiology*, Vol. 133 (2003), pp.1666-1670.
26. Gregory Graff, Susan Cullen, Kent Bradford, David Zilberman, and Alan Bennett, "The Public-Private Structure of Intellectual Property Ownership in Agricultural Biotechnology," *Nature Biotechnology*, Vol. 21, No. 9 (September 2003), pp. 989-995.
27. Gregory Graff, Gordon Rausser, and Arthur Small, "Agricultural Biotechnology's Complementary Intellectual Assets," *Review of Economics and Statistics*, Vol. 85, No. 2 (May 2003), pp. 349-363.
28. Gregory Graff, Amir Heiman, and David Zilberman, "University Research and Offices of Technology Transfer," *California Management Review*, Vol. 45, No. 1 (Fall 2002), pp. 89-115.
29. Gregory Graff and David Zilberman, "An Intellectual Property Clearinghouse for Agricultural Biotechnology," *Nature Biotechnology*, Vol. 19 (December 2001), pp. 1179-1180.

#### **Refereed Chapters in Books:**

1. Gregory Graff, David Jefferson, Monica Alandete-Saez, Cecilia Chi-Ham, Sara Boettiger, and Alan Bennett, "Intellectual Property in Agricultural Biotechnology: Strategies for Open Access," Chapter 17 in Neil Stewart, Ed., *Plant Biotechnology and Genetics: Principles, Techniques, and Applications*, 3<sup>rd</sup> Edition, 2025.

2. Gregory Graff, Benjamin Nordbrock, and Andrew Seidl, "An introduction to the economics of climate change and animal health," in C. Stephen & C. Duncan, Eds., *Climate Change and Animal Health*, Routledge, 2023.
3. Gregory D. Graff, Felipe de Figueiredo Silva, and David Zilberman, "Venture capital and the transformation of private R&D for agriculture," in *The Economics of Agricultural Research and Innovation*, P. Moser, Ed., National Bureau of Economic Research (NBER), 2021.
4. Ghulam Samad and Gregory Graff, "The urban concentration of innovation and entrepreneurship in agricultural and natural resource industries," in *Urban Studies and Entrepreneurship: How can cities foster Entrepreneurship?* M. Iftikhar, J. Justice, & D. Audretsch, Eds., Springer, 2020.
5. Gregory Graff, Gal Hochman, Chubashini Suntharlingam, and David Zilberman, "The Competing Policy Paradigms of Agricultural Biotechnology: Implications and Opportunities for Emerging and Developing Economies," in *Biotechnology for a Second Green Revolution in India*, N.C. Rao, C. Pray, & R. Herring, Eds., Academic Foundation, 2018.
6. Monica Alandete-Saez, Cecilia Chi-Ham, Gregory Graff, Sara Boettiger and Alan B. Bennett, "Intellectual Property in Agricultural Biotechnology: Strategies for Open Access" in *Plant Biotechnology and Genetics: Principles, Techniques, and Applications*, 2nd Edition, N. Stewart, Ed., Wiley: 2016, pp. 347-365.
7. Gregory Graff and David Zilberman, "How the 'Regulatory-IP' complex affects incentives to develop socially beneficial products from agricultural genomics" in E. Marden, N. Godfray, R. Manion, Eds., *Innovation in Agricultural Genomics: Overcoming Complexities in the Intellectual Property-Regulatory Complex*, University of British Columbia Press, 2015.
8. Gregory Graff, David Zilberman, and Gal Hochman, "The political economy of technology regulation in the face of creative destruction: the case of agricultural biotechnology," in R. Herring, Ed., *Handbook of Food, Politics, and Society*, Oxford University Press, 2015.
9. Monica Alandete-Saez, Cecilia Chi-Ham, Gregory Graff, and Alan Bennett, "Intellectual Property in Agriculture," in N. Van Alfen, Ed., *Encyclopedia of Agriculture and Food Systems*, Elsevier, 2014.
10. Derek Eaton and Gregory Graff, "The dynamic IP system in crop genetics and biotechnology," Chapter 28 in S.J. Smyth, P.W.B. Phillips, and D. Castle, Eds., *Handbook on Agriculture, Biotechnology, and Development*, Edward Elgar, 2013.
11. Gregory Graff, Gal Hochman, David Zilberman, "The research, development, commercialization, and adoption of drought and stress tolerant crops," in N. Tuteja and S. Gill, *Crop Improvement under Adverse Conditions*, New York: Springer Science + Business Media, 2013, pp. 1-34.
12. Gregory Graff and David Zilberman, "Agricultural Biotechnology: Equity and Prosperity," in J. Popp, M. Jahn, M. Matlock, N. Kemper, Eds. *The Role of Biotechnology in a Sustainable Food Supply*, Cambridge University Press, 2012.
13. Gregory Graff, "¿Ecos de la Ley Bayh-Dole? Un Estudio de las Políticas de PI y de Transferencia de Tecnología en las Economías Emergentes y en Desarrollo," in Anguita, Diaz, Chi-Ham, et al. Eds., *Gestion de la Propiedad Intelectual e Innovacion en Agricultura y en Salud*, Fundacion para la Innovacion Agraria (FIA) and PIPRA: Davis, CA. 2010.
14. Alan B. Bennett, Cecilia Chi-Ham, Gregory Graff, and Sarah Boettiger, "Intellectual Property in Agricultural Biotechnology: Strategies for Open Access" in N. Stewart, Ed., *Plant Biotechnology and Genetics: Principles, Techniques, and Applications*, New York: J. Wiley & Sons, 2008.
15. Gregory Graff and David Zilberman, "The Political Economy of Intellectual Property: Re-Examining European Policy on Plant Biotechnology" in J. Kesan, Ed., *Seeds of Change: Intellectual Property Protection for Agricultural Biotechnologies*, Wallingford, UK: CABI Press, 2007.

16. Karl Bergman and Gregory Graff, “Collaborative IP management for stem cell R&D: Lessons learned from agricultural biotechnology” in *Intellectual Property and Bioscience*, Stuttgart/Berlin: Steinbeis-Edition, 2007.
17. Gregory Graff, “Echoes of Bayh-Dole: A Survey of Intellectual Property and Technology Transfer Policies in Emerging and Developing Economies” in A. Krattiger, R. Mahoney, L. Nelsen, et al, Eds., *Intellectual Property Management in Health and Agricultural Innovation: A Handbook of Best Practices*, MIHR: Oxford, UK and PIPRA: Davis, CA. 2007.
18. Martha Mutschler and Gregory Graff, “Introduction to Intellectual Property Issues in the University Setting: A Primer for Scientists” in A. Krattiger, R. Mahoney, L. Nelsen, et al, Eds., *Intellectual Property Management in Health and Agricultural Innovation: A Handbook of Best Practices*, MIHR: Oxford, UK and PIPRA: Davis, CA. 2007.
19. Gregory Graff and Alan Bennett, “Intellectual Property and Technology Transfer by the University of California Agricultural Experiment Station” in A. Krattiger, R. Mahoney, L. Nelsen, et al, Eds., *Intellectual Property Management in Health and Agricultural Innovation: A Handbook of Best Practices*, MIHR: Oxford, UK and PIPRA: Davis, CA. 2007.
20. Gregory Graff, “Regulation of Transgenic Crops Intended for Pharmaceutical and Industrial Uses” in J. Alston, R. Just, and D. Zilberman, Eds., *Regulating Agricultural Biotechnology: Economics and Policy*, New York: Springer, 2006.
21. Gregory Graff, David Roland-Holst, and David Zilberman, “Agricultural Biotechnology and Globalization: U.S. Experience with Public and Private Sector Research” in J.H.H. Wesseler, Ed., *Environmental Costs and Benefits of Transgenic Crops*, Dordrecht: Springer, 2005.
22. Gregory Graff and David Zilberman, “Towards an Intellectual Property Clearinghouse for Agricultural Biotechnology,” in J. Cooper, L. Lipper, and D. Zilberman, Eds., *Agricultural Biodiversity and Biotechnology in Economic Development*, New York: Springer, 2005.
23. Alain de Janvry, Gregory Graff, Elisabeth Sadoulet, and David Zilberman, “Technological Change in Agriculture and Poverty Reduction: The Potential Role of Biotechnology” in J. Cooper, L. Lipper, and D. Zilberman, Eds., *Agricultural Biodiversity and Biotechnology in Economic Development*, New York: Springer, 2005.
24. Gregory Graff, Matin Qaim, Cherisa Yarkin, and David Zilberman, “Agricultural Biotechnology in Developing Countries,” in C.G. Scanes and J.A. Miranowski, Eds., *Perspectives in World Food and Agriculture*, Ames, Iowa: Iowa State Press, 2004.
25. Sara Boettiger, Gregory Graff, Philip Pardey, Eric Van Dusen, and Brian Wright, “Intellectual Property Rights for Plant Biotechnology: International Aspects,” in P. Christou and H. Klee, Eds., *The Handbook of Plant Biotechnology*, Chichester: John Wiley & Sons Ltd., 2004.
26. Gregory Gaff, Amir Heiman, David Zilberman, Federico Castillo, Douglas Parker, “Universities, Technology Transfer, and Industrial R&D,” in R.E. Evenson, V. Santaniello, and D. Zilberman, Eds., *Economic and Social Issues in Agricultural Biotechnology*, Wallingford: CABI, 2002.

### **Refereed Proceedings/Transactions**

1. Gregory D. Graff, Charles deGrazia, and Nicholas Rada, *The Contribution of Startups, Venture Finance, and Patenting to Innovation in U.S. Agriculture*. USPTO Economic Working Paper No. 2024-2, U.S. Patent and Trademark Office, Alexandria, VA. July 2024. Available at SSRN: <https://ssrn.com/abstract=4892791>.
2. Intan Hamdan-Livramento, Gregory D. Graff, and Alica Daly, *Innovation Complexity in AgTech: The case of Brazil, Kenya and the United States of America*. Economic Research Working Paper Series



No. 82, World Intellectual Property Organization (WIPO), Geneva, Switzerland. April 2024.  
Available at SSRN: <https://ssrn.com/abstract=4816054>.

3. Gregory D. Graff and Intan Hamdan-Livramento, *Global Roots of Innovation in Plant Biotechnology*, Economic Research Working Paper No. 59, World Intellectual Property Organization (WIPO), Geneva, Switzerland. November 2019.
4. Per Hulthén and Gregory D. Graff, *Migration patterns of venture capital funded startups*, Department of Technology Management and Economics, Chalmers University of Technology, Gothenburg, Sweden. 2019.
5. Chan, Sarah, Gregory Graff, Kazuto Kabo, Debra J.H. Mathews, Alan Regenberg, Douglas Sipp, and David Winickoff. *Statement on Data and Materials Sharing and Intellectual Property in Pluripotent Stem Cell Science in Japan and China*. Hinxton Group Asia, Kobe, Japan, January 2012.
6. Chan, Sarah, Peter J Donovan, Ruth Faden, John Harris, Robin Lovell-Badge, Debra JH Mathews, Alan Regenberg, Julian Savulescu, David Winickoff, Robert Cook-Deegan, Gregory Graff, Aurora Plomer, Kris Saha, Christopher Scott, John Sulston, and Patrick Taylor. *Statement on Policies and Practices Governing Data and Materials Sharing and Intellectual Property in Stem Cell Science*. Hinxton Group, November 2010.
7. Sean O'Connor, Gregory Graff, and David Winickoff, *Legal Context of University Intellectual Property and Technology Transfer*, National Academies of Science, Science Technology Economics and Policy (STEP) Board, September 2010.

#### **Non-Refereed Journal Articles/Chapters/Proceedings/Transactions:**

1. Gregory Graff, Karl Bergman, Alan Bennett, and David Zilberman, "Intellectual Property Clearinghouses as an Institutional Response to the Privatization of Innovation in Agriculture," *African Technology Development Foundation Journal*, Vol. 3, No. 3, Fall 2006.
2. Gregory Graff and GianCarlo Moschini, "Pharmaceutical and Industrial Products in Crops: Economic Prospects and Impacts," *Iowa Ag Review*, Vol. 10, No. 4, Fall 2004, pp. 4-5, 11.
3. Gregory Graff and David Zilberman, "Explaining Europe's Resistance to Agricultural Biotechnology," *Agricultural and Resource Economics Update*, Vol. 7, No. 5, May 2004, pp. 1-4.
4. Gregory Graff, Amir Heiman, Cherisa Yarkin, and David Zilberman, "Privatization and Innovation in Agricultural Biotechnology," *Agricultural and Resource Economics Update*, Vol. 6, No. 4, March-April 2003, pp. 5-7.
5. Gregory Graff and David Zilberman. "Towards an Intellectual Property Clearinghouse for Agricultural Biotechnology," *IP Strategy Today*, No. 3, 2001, pp. 1-13.

#### **Other (e.g. lab texts, book reviews, technical reports, in-house reports):**

1. Per Hulthen and Gregory Graff, *Exit Value Study Third Stage Research Project: Summary Report*, Chalmers University of Technology, Gothenburg, Sweden. November 2024.
2. Dawn Thilmany, Becca Jablonski, and Gregory Graff, *The Colorado Blueprint of Agriculture and Food*, Colorado State University, Fort Collins, CO, 2018.
3. Gregory Graff, Annabelle Berklund, and Kathay Rennels, *The Emergence of an Innovation Cluster in the Agricultural Value Chain Along the Colorado Front Range*, Colorado State University, Fort Collins, CO, November 2014.
4. Patrick Byrne, Dustin Pendell, and Gregory Graff, "Labelling of Genetically Modified Foods," Fact Sheet No. 9.371, Colorado State University Extension, Fort Collins, CO, October 2014.

5. Gregory Graff, Ryan Mortenson, Rebecca Goldbach, et al, *The Value Chain of Colorado Agriculture*, Department of Agricultural and Resource Economics and the Office of Engagement, Colorado State University, Fort Collins, CO, February 2013.
6. Jennifer Bond and Gregory Graff, book review of “G. Alsos, S. Carter, E. Ljunggren, and F. Welter (eds.), *The Handbook of Research on Entrepreneurship in Agriculture and Rural Development*. Northampton: Edward Elgar Publishing, Inc. 320 pp., (2012)” in *Agribusiness*, Volume 28, Issue 1, Winter 2012, pp. 118–120.
7. Gregory Graff, David Zilberman, and Alan Bennett, *Nutritional and Product Quality Innovations in Agricultural Biotechnology: A new generation of crop traits and their potential economic impacts*, Public Intellectual Property Resource for Agriculture (PIPRA), Davis, CA (April 2008), 142 pages.
8. Karl Bergman and Gregory Graff, *Collaborative IP Management for Stem Cell Research and Development*, Center of Intellectual Property Studies (CIP), Gothenberg, Sweden, and the Public Intellectual Property Resource for Agriculture (PIPRA), Davis, CA (December 2006), 46 pages.
9. Gregory Graff, *The Impact of Blocking Patents on Innovation in Soy Genetics*, report for the United Soybean Board/American Soybean Association, St. Louis, MO (July 2005), 62 pages.
10. Ann Bublitz and Gregory Graff, *Crop Biomanufacturing Part II: Implications for the Farm Sector*, Bio Economic Research Associates (bio-era), Cambridge, MA (January 2004), 14 pages.
11. Gregory Graff, *Crop Biomanufacturing Part I: Economic Opportunities for the Biotechnology Industry*, Bio Economic Research Associates (bio-era), Cambridge, MA (January 2004), 36 pages.
12. Gregory Graff and James Newcomb, *Agricultural Biotechnology at the Crossroads: the Changing Structure of Industry*, Bio Economic Research Associates (bio-era), Cambridge, MA (February 2003), 26 pages.

## CONTRACTS & GRANTS

### Externally-Funded Projects as PI

|             |                                                                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 – 2023 | Research Intergovernmental Personnel Agreement (IPA) one year extension, United States Patent and Trademark Office, \$38,251.                                                                                           |
| 2020 – 2022 | Research Intergovernmental Personnel Agreement (IPA), United States Patent and Trademark Office, \$82,680.                                                                                                              |
| 2019 – 2020 | Venture capital funded agricultural innovation in the U.S. and abroad, USDA-ERS-Economic Research Service Cooperative Agreement, \$23,973.                                                                              |
| 2019 - 2021 | Intellectual Property on CRISPR and the Gene Editing of Crops for Sustainable Agriculture, USDA, National Institutes of Food and Agriculture (NIFA), \$44,887.                                                          |
| 2018 - 2019 | Intellectual Property and Access to CRISPR-Cas Gene Editing Technologies for Innovation in Crop Agriculture, OECD, Co-operative Research Programme (CRP), Trade and Agriculture Directorate, €28,215.                   |
| 2018 - 2020 | Venture Capital and the Transformation of Private R&D for Agriculture and Food, National Bureau of Economic Research (NBER), joint with David Zilberman, University of California Berkeley, \$6,000.                    |
| 2013 - 2014 | Incentives for Agricultural Innovation in Emerging and Developing Economies: Understanding National Intellectual Property Policies in a Global Context, International Food Policy Research Institute (IFPRI), \$15,274. |



## Externally-Funded Projects as CoPI

### *Proposal submitted*

2026 - 2029    The Roles of Startups, Corporations, and Public Institutions in U.S. Agricultural Innovation, with Ruiqing Miao and Jiaqi Wang, Auburn University, and Diane Charlton, Montana State University. U.S. Department of Agriculture, USDA-NIFA-AFRI-011134. Proposal submitted October 2025. Funds requested for Colorado State University \$219,881 out of total \$800,000.

### *Projects completed*

2017 - 2021    Root Genetics in the Field to Understand Drought Adaptation and Carbon Sequestration, with PI John McKay, co-PI on Technology to Market, ARPA-E, U.S. Department of Energy, \$8.3 million. (Interdisciplinary between crop science and agricultural economics; role as director of “Technology to Market” strategy and economic research on potential market uptake of crops with improved root traits.)

2012 - 2013    The Value Chain of Colorado Agriculture, with co-PI Kathay Rennels (CSU Office of Engagement), Colorado Department of Agriculture, State of Colorado, \$50,000. (Interdisciplinary; lead data compilation and authorship of report on the structure of the state’s agricultural and food industry; informing economic development strategy of the Colorado Office of Economic Development and International Trade and the Governor of Colorado)

2009 - 2013    International Assessments of Patenting in Genetics, with Co-PI Philip Pardey (University of Minnesota), National Institutes of Health, \$1.4 million. (Interdisciplinary between agricultural economics, intellectual property law, public policy, and biology; as Co-PI, I led the data collection and data analysis efforts, and authorship of main publications.)

2009 - 2011    Technological and Agricultural Entrepreneurship for a Globally Sustainable Future, with co-PIs Anthony Marchese (CSU, Mechanical Engineering) and Paul Hudnut (CSU, Management), National Collegiate Inventors and Innovators Association (NCIIA), \$49,900. (Interdisciplinary course development project between engineering, business, and agricultural economics; each co-PI shared equally in experimental course development and teaching load for the two semesters that the experimental course was offered.)

2010 - 2011    Designing a Technology-Neutral Benefit Pricing Policy for the Electric Power Sector in Colorado, with Co-PIs Terry Iverson (CSU) and Catherine Keske (CSU), Governor’s Energy Office, State of Colorado, \$93,177. (Disciplinary; I advised on strategy and content, with only a minor role in writing.)

2009 - 2010    Legal Context of University Intellectual Property (IP) and Technology Transfer, with Co-PIs Sean O’Connor (University of Washington) and David Winickoff (University of California Berkeley), National Academies of Science, Board on Science, Technology, and Economic Policy (STEP) and Committee on Science, Technology, and Law (CSTL), \$4,500. (Interdisciplinary between economics, policy, and law; played supporting role in research and authoring report.)

2007 - 2010 Intellectual Property Protection and Contractual Relations for Biofuels Innovations, with PI Brian Wright (University of California Berkeley), Energy Bioscience Institute (EBI), \$318,000.

#### **Externally-Funded Projects as Investigator or role other than PI or CoPI**

2020 – 2024 Exit Value Study, Third Stage, with PI Mats Lundqvist (Chalmers University of Technology, Gothenburg, Sweden) and project manager Per Hulten (Chalmers University of Technology, Gothenburg Sweden), external collaborator and subcontractor, Vinnova, Carl Bennet AB, Region of Western Sweden (VGR), SEK5.2 million.

2013 - 2016 PACE-'Omics: Personalized, Accessible, Cost-Effective applications of 'Omics technologies, with PIs Christopher McCabe (University of Alberta) and Tania Bubela (University of Alberta), external collaborator, Genome Canada, CAD\$8.9 million. (Interdisciplinary between biology, public policy, and economics; played minor advisory role, attended and presented at workshops.)

2009 - 2011 Harnessing the Sun for On-farm Fertilizer Production, with PI Jessica Davis (CSU), Western Sustainable Agricultural Research and Extension (WSARE), US Department of Agriculture, \$159,023. (Interdisciplinary between soil science, microbiology, and agricultural economics; played supporting role analyzing market potential and advised a graduate student team recruited from the Global Social and Sustainable (GSSE) MBA program in the College of Business assigned to develop a marketing strategy for the technology.)

#### **Internally-Funded Awards**

2021-2025 Agribusiness and Food Innovation Management master's program startup funds, CSU System Chancellor's Office

2020 - 2023 Soil Carbon Solutions Center (SCSC), Center for Innovative Partnerships (CIP) initiative of CSU Vice President for Research, with PI Keith Paustian.

2017 - 2018 Strategic Insights into the Regional Network of Agricultural Technology Innovators in Colorado, CSU Vice President for Research Quarterly Funding, CSU Ventures, College of Agricultural Sciences, Department of Agricultural and Resource Economics.

2016 - 2017 Colorado Food Blue Print: Opportunities for Colorado's Food System, with Co-PIs Dawn Thilmany and Becca Jablonski, CSU College of Agricultural Sciences, \$116,000.

2015 - 2017 Innovation Center for Sustainable Agriculture (ICSA), Center for Innovative Partnerships (CIP) initiative of CSU Vice President for Research, with PI Matt Wallenstein.

2013 - 2014 Characterizing Colorado's Emergent Agricultural Innovation Cluster, with co-PI Kathay Rennels, CSU Chancellor's Venture Fund, CSU Ventures, and CSU College of Agricultural Sciences, total project amount \$19,625.

2008 - 2009 The Intellectual Property Landscape of Alternative Biofuel Technologies, Clean Energy SuperCluster Internal Grant, \$14,775.

### **Un-Funded Projects as PI or CoPI**

|             |                                                                                                                                                                                                                                                                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2020        | Creating a Masters Program in Agribusiness Entrepreneurship and Innovation Management: Higher Education in Creative Partnerships with the Business Sector, USDA-NIFA, Higher Education Challenge, \$150,000.                                                                                                                                    |
| 2020        | Venture Capital Investment in Agricultural Innovation and Publicly Funded Agricultural Science, USDA-NIFA-National Institute of Food and Agriculture, January 1, 2021 - December 31, 2023                                                                                                                                                       |
| 2019        | Creating a Masters Program in Agribusiness Entrepreneurship and Innovation Management: Higher Education in Creative Partnership with the Business Sector, USDA-NIFA, Higher Education Challenge, \$150,000.                                                                                                                                     |
| 2018        | Creating a Masters Program in Agribusiness Entrepreneurship and Innovation Management: Higher Education in Creative Partnership with the Business Sector, USDA-NIFA, Higher Education Challenge, \$150,000.                                                                                                                                     |
| 2017 - 2021 | Putting Waste to Work: Enabling Adoption of Organic Soil Amendments for Improved Agricultural Water Use Efficiency (126240), with PI Rich Conant (CSU), National Science Foundation (INFEWS/T3), \$3.0 million.                                                                                                                                 |
| 2016        | Economic Potential of Industrial Hemp in the U.S., Industrial Hemp Research Foundation, \$245,700.                                                                                                                                                                                                                                              |
| 2015        | The Economics of Universities and Intellectual Property in the Knowledge Economy, Mobility for Regional Excellence, Vastra Gotaland Region, Sweden, 1.6 million Swedish krona (SEK)                                                                                                                                                             |
| 2015        | The role of the research university in fostering commercial innovation and entrepreneurship, with co-PI Mats Lundqvist (Chalmers University of Technology), Ewing Marion Kauffman Foundation, \$65,000.                                                                                                                                         |
| 2014        | Universities and Intellectual Property in the Knowledge Economy and Innovation Management by Student Entrepreneurs, Fulbright Scholar Program, 2015-16 U.S. Scholar Application, \$22,950.                                                                                                                                                      |
| 2013        | Efficacy of translating genomic research to innovations in a global IP environment, with Co-PI Philip Pardey (University of Minnesota), National Institutes of Health, \$1.9 million.                                                                                                                                                           |
| 2013        | Universities and Intellectual Property in the Knowledge Economy and Innovation Management by Student Entrepreneurs, Fulbright Scholar Program, 2014-15 U.S. Scholar Application, \$22,950.                                                                                                                                                      |
| 2012        | Integrating Bibliometrics and Econometrics to Evaluate the Economic Consequences of Research: with Application to the Food and Agricultural Sciences, with Co-PIs Matt Anderson (University of Wyoming) and Philip Pardey (University of Minnesota), Science of Science and Innovation Policy (SciSIP), National Science Foundation, \$787,714. |

### **CONFERENCES/MEETINGS ORGANIZED**

- Impact Analyses and Decision Strategies for Agricultural Research, NC1034 Multistate Research Conference and Annual Meeting, University of California Berkeley, March 29-30, 2025.
- Impact Analyses and Decision Strategies for Agricultural Research, NC1034 Multistate Research Conference and Annual Meeting, Colorado State University, March 29-30, 2024.
- Impact Analyses and Decision Strategies for Agricultural Research, NC1034 Multistate Research Conference and Annual Meeting, Virginia Tech Executive Briefing Center, Arlington VA, March 30-31, 2023.
- *Nutrition Security and Resilience Through Ag Innovation*, NC1034 Multistate Research Conference, Anaheim, CA, July 30, 2022.
- *Advances in Agricultural Technology, Productivity, and Sustainability*, NC1034 Multistate Research Conference, (hosted on ZOOM), December 3-4, 2021.
- *Control and Access: Intellectual property and CRISPR-Cas gene editing for innovation in crop agriculture*, sponsored by OECD and USDA-NIFA, Keystone, CO, October 23-24, 2019
- *CSU Ag Innovation Summit*, with College of Agricultural Sciences, Lory Student Center, Colorado State University, September 6-7, 2017.
- *CSU Ag Innovation Summit: Advancing the Agricultural Economy through Innovation*, in collaboration with College of Agricultural Sciences, CSU Office of Engagement, and the Van Heyst Group, Lory Student Center, Colorado State University, March 18-20, 2015.
- *Analyzing Policy Impacts on Biotechnology Using Patent Data*, Rayburn Office Building, U.S. House of Representatives, Washington DC March 13-14, 2014. Organized briefing workshop in collaboration with colleagues at Duke University, McGill University, and University of Alberta.
- *The Venture Engine: How a School of Entrepreneurship can Bridge the Gap from Invention to Marketplace*, Colorado State University, August 15, 2008.
- *Institutional Landscapes in Stem Cell Research and Development: Problems and Solutions*, sponsored by Berkeley Stem Cell Center and UC Berkeley Science, Technology, and Society Center, February 6, 2008, Mission Bay Conference Center, San Francisco, CA.
- *Information Technology in Agriculture, Resources, and the Environment: Linking University Research and Commercial Application*, UC Berkeley, May 27, 2004.
- *Sharing the Flame: Excellence in Licensing Practices*, 2003 Winter Meeting, Licensing Executives' Society, February 12-15, 2003, Salt Lake City. Member of conference organizing committee.
- *Intellectual Property Clearinghouse Mechanisms for Agriculture*, UC Berkeley, February 16, 2001.

## PAPERS PRESENTED/SYMPOSIA/INVITED LECTURES/PROFESSIONAL MEETINGS/WORKSHOPS

1. Gregory Graff, "Advancing agricultural technology and innovation," invited Plenary Panel speaker, Fall 2025 Fulbright Visiting Scholars Seminar, *America250: Tradition, Technology, and Innovation in the Mountain West*, Colorado State University, December 10, 2025.
2. Gregory Graff, Charles deGrazia, Nicholas Rada, "The contribution of startups, venture finance, and patents to innovation in U.S. agriculture," International Consortium for Applied Bioeconomy Research (ICABR) annual conference, Saskatoon, Saskatchewan, Canada, July 8-10, 2025.
3. Gregory Graff, Charles deGrazia, Nicholas Rada, "The contribution of startups, venture finance, and patents to innovation in U.S. agriculture," NC1034 Annual Meeting, University of California, Berkeley, March 28-29, 2025.
4. Gregory Graff, Charles deGrazia, Nicholas Rada, "Startups, private finance, and innovation in U.S. agriculture," 18th Annual Conference of the European Policy for Intellectual Property Association (EPIP) 2023, Kraków, Poland, September 11-13, 2023.
5. Gregory Graff, Charles deGrazia, Nicholas Rada, "Startups, private finance, and innovation in U.S. agriculture," invited presentation at Agricultural Productivity Growth: Measurement, Drivers, and

- Climatic Effects, USDA-ERS, Farm Foundation, GAP Initiative-CALS Global, Virginia Tech, Executive Briefing Center (EBC), Arlington, VA, March 29-30, 2023.
6. Gregory Graff, Charles deGrazia, Nicholas Rada, "The Scope of Technological Innovations Funded by Venture Capital in the Agri-Food Sectors," 17<sup>th</sup> Annual Conference of European Policy for Intellectual Property, September 14-16, 2022.
  7. Gregory Graff, Charles deGrazia, Nicholas Rada, "The Scope of Technological Innovations Funded by Venture Capital in the Agri-Food Sectors," AAEA Organized Symposia, August 23, 2022.
  8. Gregory Graff, Charles deGrazia, Nicholas Rada, "The Scope of Technological Innovations Funded by Venture Capital in the Agri-Food Sectors," NC1034 Annual Research Conference, Anaheim, CA July 31, 2022.
  9. Gregory Graff, "Models of Technology Transfer for Genome-Editing Technologies" invited presentation, Plant & Animal Genome Conference: PAG 30, San Diego, CA, January 13-18, 2022.
  10. Gregory Graff, "The Role of Technology and Data" discussant, Agricultural Symposium on *The Roots of Agricultural Productivity Growth*, Kansas City Federal Reserve Bank, May 24-25, 2021.
  11. Gregory Graff, Felipe De Figueiredo Silva, and David Zilberman, "Venture Capital and the Transformation of Private R&D for Agriculture and Food," invited departmental seminar, Agricultural and Resource Economics, North Dakota State University, Fargo, ND (via ZOOM), January 22, 2021.
  12. Gregory Graff, "Intersections of Venture Capital, Intellectual Property, and Innovation in Agriculture," invited presentation for the Intellectual Property Committee of the American Seed Trade Association (ASTA), (via ZOOM) January 14, 2021.
  13. Gregory Graff, "Intellectual Property as a Governance Mechanism for Genome Editing in Agriculture and Food," invited presentation for conference on Gene Editing in Agriculture and Food: Social Concerns, Public Engagement and Governance, Iowa State University, October 20-21, 2020.
  14. Intan Hamdan-Livramento and Gregory Graff, "Global Roots of Innovation in Plant Biotechnology," presentation at ICABR Virtual Conference, October 15, 2020.
  15. Gregory Graff, Felipe De Figueiredo Silva, and David Zilberman, "Venture Capital and the Transformation of Private R&D for Agriculture and Food," invited presentation for session on "Studying Innovation in Agriculture – New data and tools," ASSA, San Diego, January 3, 2020.
  16. Gregory Graff, "Intellectual property and CRISPR-Cas gene editing for innovation in crop agriculture," invited seminar, Innovative Genomics Institute, University of California Berkeley, December 3, 2019.
  17. Gregory Graff, "Innovation and the Value Chain of Agriculture and Food," invited talk for Colorado Farm Bureau 2019 Annual Meeting, Denver, CO, November 22, 2019.
  18. Gregory Graff, "The CRISPR patent race: Intellectual property and CRISPR-Cas gene editing for innovation in crop agriculture," invited video keynote lecture for Genetics 4.0 - CRISPR Revolution / Genética 4.0 - Revolução CRISPR, Escola Superior de Agricultura Luiz de Queiroz, University of São Paulo, Piracicaba, Brasil, 13 November 2019.
  19. Gregory Graff, "Control and Access: Intellectual property and CRISPR-Cas gene editing for innovation in crop agriculture," invited keynote address 2nd Annual Genome Editing Symposium, Texas A&M University, 3 October 2019.
  20. Gregory Graff, "The Economics of CRISPR: Factors governing the application of gene editing in the agricultural value chain," ARBL/Physiology/Biomedical Sciences seminar, College of Veterinary Medicine, Colorado State University, September 23, 2019.
  21. Gregory Graff, "Trends in Innovation and the Value Chain of Agriculture and Food," invited panel presentation, German/American Chamber of Congress Agritech Symposium, Denver, CO, September 19, 2019.

22. Gregory Graff, “Trends in Innovation and the Value Chain of Agriculture and Food,” invited keynote address for the Colorado Water Congress 2019 Summer Conference, Steamboat Springs, CO, August 20-22, 2019.
23. Gregory Graff, “Venture Capital and the Transformation of Private R&D for Agriculture and Food,” invited keynote lecture for International Association of Milk Control Agencies, 2019 Annual Meeting, Golden, Colorado, August 14, 2019.
24. Gregory Graff, Felipe De Figueiredo Silva, and David Zilberman, “Venture Capital and the Transformation of Private R&D for Agriculture and Food,” invited chapter for *Economics of Agricultural Research and Innovation*, National Bureau of Economic Research (NBER), Washington DC, May 17, 2019
25. Gregory Graff, “Controlling CRISPR: Intellectual property and governance of the collaborative technology platform for genome editing,” invited presentation to the Working Group on Bio-, Nano-, and Converging Technologies, Organisation for Economic Co-operation and Development (OECD), Paris, 16 May 2019.
26. Gregory Graff, Felipe De Figueiredo Silva, and David Zilberman, “Venture Capital Funding for Agricultural Innovation,” Innovation in Agrifood Supply Chains: Finance, Profitability, and Sustainability, Solidaridad and University of California Berkeley’ Berkeley, CA, April 10-11, 2019.
27. Gregory Graff, “Innovation networks and clusters in agricultural biotechnology,” invited presentation for workshop on *Geography of Innovation: Local Hotspots, Global Networks*, World Intellectual Property Organization (WIPO), Geneva, Switzerland, February 25, 2019.
28. Gregory Graff and Ghulam Samad, “The Urban Concentration of Innovation and Entrepreneurship in Agricultural and Natural Resource Industries,” 65<sup>th</sup> Annual North American Meetings of the Regional Science Association International (RSAI), North American Regional Science Council (NARSC), San Antonio, TX, November 7, 2018
29. Gregory Graff, Felipe De Figueiredo Silva, and David Zilberman, “Venture Capital and the Transformation of Private R&D for Agriculture and Food,” Economics of Agricultural Research and Innovation, National Bureau of Economic Research (NBER), Cambridge, MA, November 8, 2018
30. Gregory Graff, “IP Control of CRISPR: Implications for Development of Agricultural and Food Applications,” Pioneer Policy Lecture Series, Iowa State University, August 27, 2018.
31. Gregory Graff, “IP Control of CRISPR: Implications for Development of Agricultural and Food Applications,” Presidential Invited Paper, AAEA Annual Meetings, Washington DC, August 6, 2018.
32. Ghulam Samad and Gregory Graff, “The Urban Clustering of Innovation for Rural Industries: Biotechnologies for Agriculture, Energy, and Natural Resources” Western Agricultural Economics Association (WAEA) Annual Meetings, Anchorage, AK, June 20, 2018.
33. Gregory Graff, “The Economics of CRISPR: Factors governing the application of gene editing in the agricultural value chain,” International Consortium on Applied Bio-Economy Research (ICABR), The World Bank, Washington DC, June 12-15, 2018.
34. Gregory Graff, “Energy and Agriculture: How dependence of agriculture on energy inputs poses the biggest opportunities for innovation,” UC Davis Innovation Summit, Innovation Institute for Food and Health, University of California Davis, April 23, 2018.
35. Gregory Graff, “Controlling CRISPR: Factors governing the application of gene editing in the agricultural value chain,” *Innovation in Agrifood Supply Chains: People-Planet-Profit*, University of California Berkeley, April 18-19, 2018.
36. Yoo Hwan Lee and Gregory Graff, “Empirical Estimation of University Knowledge Production Functions for Knowledge Outputs Disseminated via Multiple Channels,” ASSA Annual Meetings, Philadelphia, January 7, 2018

37. Gregory Graff, "Entrepreneurship in Agricultural Technology," invited panel presentation for symposium on *Research and Innovation Policies for Sustainable Productivity Growth in Agriculture*, Farm Foundation, USDA-ERS, and OECD, The National Press Club, Washington, DC, September 19-20, 2017.
38. Gregory Graff and David Zilberman, "Intellectual Property and Regulatory Issues of Gene Editing in Agriculture and Food: Lessons of GMOs for CRISPR Gene Editing," 2017 AAEA Annual Meeting, Chicago, IL, August 1, 2017
39. Gregory Graff, "Controlling CRISPR: How the rapidly changing patent and regulatory environment may affect commercial applications of genome editing," invited keynote address for Rocky Mountain Food Safety Conference, Denver, CO, May 24, 2017.
40. Gregory Graff, "Innovation and Regulatory Impact of Gene Modification vs. Gene Editing of Food and Crops" invited panel presentation at conference Planting the Seed 2017, hosted by Barnes and Thornburg LLP, Indianapolis, IN, May 23 2017.
41. Gregory Graff and David Zilberman, "Not Your Mother's GMOs: Intellectual Property and Regulatory Issues of Gene Editing in Agriculture and Food," Presented at NC-1034 conference Impact Analyses and Decision Strategies for Agricultural Research, University of Arizona, Tucson, AZ, February 25, 2017.
42. Gregory Graff, "The Rise and Fall of Innovation in Biofuels: the global landscape of biofuel patenting from 1970 to 2013" invited paper presentation at conference The Bioeconomy: Technological and Policy Path Forward, Rutgers University, New Brunswick, NJ, October 1, 2016.
43. Gregory Graff, "The Emerging Patent Landscape of CRISPR-Cas Gene Editing Technology," comments presented at the Workshop on Gene Editing in an International Context: Scientific, Economic, and Social Issues across Sectors, Working Party on Biotechnology, Nanotechnology, and Converging Technologies, OECD, Ottawa, Canada, September 29-30, 2016.
44. Gregory Graff, "The Emerging Patent Landscape of CRISPR-Cas Gene Editing Technology," invited panel presentation at the CIP Forum 2016, Center for Intellectual Property (CIP), University of Gothenburg, Gothenburg, Sweden, September 27 2016.
45. Jakrapun Suksawat & Gregory Graff, "Potential Impact of Cropping-pattern and Land-use Changes from the Thai Ethanol Production Mandate on the Environment" selected paper presentation at the East Asian Association of Environmental and Resource Economics 2016 Congress (EAAERE 2016), August 7-10, 2016, Kyushu Sangyo University, Fukuoka, Japan.
46. Gregory Graff, "The Rise and Fall of Innovation in Biofuels: the global landscape of biofuel patenting from 1970 to 2013" invited seminar at Department of Economics and Management of Enterprises, Faculty of Economic Sciences, Warsaw University of Life Sciences (SGGW), Warsaw, Poland, April 7, 2016
47. Gregory Graff, "Patenting in Sub-Saharan Africa and implications for knowledge-driven economic development" invited seminar, College of Law, University of New Hampshire, March 25 2016.
48. Gregory Graff and Yoo Hwan Lee, "The Inner Workings of University Knowledge Production and its Impacts on Commercial Innovation: Empirical Evidence from Colorado State University 1989-2012," Technology Management and Economics Seminar, Chalmers University of Technology, Gothenburg, Sweden, December 3, 2015.
49. Gregory Graff, speaker for panel on "How Will We Nourish our Planet in the Face of Projected Population Growth?" Global Grand Challenges: CSU Research and Translation," National Press Club, Washington, D.C., June 11, 2015.
50. Yoo Hwan Lee and Gregory Graff, "University Research Productivity and its Impact on the Regional Agricultural Economy: The Case of Colorado State University and the Colorado Economy," invited poster presentation, 2015 AAEA/WAEA Annual Meetings, San Francisco, CA, July 27, 2015.



51. Gregory Graff, Devon Phillips, Philip Pardey, "Innovation in the Bioeconomy," DARE Research Seminar, March 26, 2015.
52. Gregory Graff, "The Business of Sustainably Feeding a World of 9 Billion," invited presentation for panel on The Business of Feeding the World, 2014 International Colloquium, Visions of Future Earth: Linking Society, Economics, and the Environment, Lory Student Center, Colorado State University, October 6-8, 2014
53. Gregory Graff, Devon Phillips, Philip Pardey, "Innovation in the Bioeconomy," invited paper presentation for Australian Agricultural & Resource Economics Society (AARES) Organized Symposium, "Innovation in the Agricultural Bioeconomy: Implications for Yield, Market Concentration, and Regulations in the Crop Sector," at the 2014 AAEA Annual Meetings, Minneapolis, MN, July 28, 2014.
54. Gregory Graff, Devon Phillips, Philip Pardey, "Innovation in the Bioeconomy," paper presented at NC1034 conference "Pests, Germs and Seeds: The Economics of Policies, Programs, and Technologies for Managing Agricultural Pests and Diseases," University of California Davis, March 28, 2014.
55. Gregory Graff, "The Impact of the U.S. Supreme Court's AMP v. Myriad decision across industries: The changing landscape of U.S. gene patents," invited presentation for Congressional Advisory Workshop "The Future of Gene Patents: Making Sense of the Supreme Court's Decision in the Myriad Case," Rayburn House of Representatives Office Building, Washington DC, March 13, 2014.
56. Gregory Graff, Devon Phillips, Philip Pardey, "Innovation in the Bioeconomy," invited papers presentation for Australian Agricultural & Resource Economics Society (AARES) Annual Meetings, Port Macquarie, NSW, Australia, 7 February 2014.
57. Gregory Graff, Devon Phillips, Philip Pardey, et al, "Not Quite a Myriad of Gene Patents: The Changing Landscape of U.S. Patents that Claim Isolated Nucleic Acids," NC1034 Conference, Biotechnology Trade and Transfer: Barriers and Opportunities, University of Arizona, Tucson, AZ, March 15, 2012.
58. Gregory Graff, Devon Phillips, Philip Pardey, et al, "Global Patenting Trends in the Genetic Sciences 1970-2010: A First Look at the INSTEPP Global Genetics Patent Database," Canadian Science Policy Conference, Calgary, Alberta, Canada, November 6, 2012.
59. Gregory Graff, Cecilia Chi-Ham, Mark Szczerba, and Alan Bennett, "The Emergence of Agbiogenetics: Implications for the Future of Crop Genetic Improvement," invited presentation at the American Agricultural Law Association (AALA) 2012 Annual Meetings, Nashville, TN, October 20, 2012.

## TEACHING

| Year | Semester | Course No./Title                          | Cr. Hrs. | Enrollment |
|------|----------|-------------------------------------------|----------|------------|
| 2025 | Fall     | AREC 586C New Venture Launch Practicum    | 5        | 2          |
| 2025 | Fall     | AREC 517 Entrepreneurial Identity & Teams | 2        | 1          |
| 2025 | Fall     | AREC 513 Idea Evaluation                  | 2        | 1          |
| 2025 | Summer   | AREC 586B New Venture Launch Practicum    | 2        | 2          |
| 2025 | Summer   | AREC 520 Intel. Prop. in Food and Ag.     | 2        | 3          |
| 2025 | Spring   | AREC 586A New Venture Launch Practicum    | 2        | 2          |
| 2025 | Spring   | AREC 518 Access to Capital in Ag. Sector  | 2*       | 3          |
| 2025 | Spring   | AREC 478 Agricultural Policy              | 3        | 32         |
| 2024 |          | Total                                     | 19       |            |

|      |             |                                             |    |    |
|------|-------------|---------------------------------------------|----|----|
| 2024 | Fall        | AREC 586C New Venture Launch Practicum†     | 5  | 11 |
| 2024 | Fall        | AREC 517 Entrepreneurial Identity & Teams   | 2  | 2  |
| 2024 | Fall        | AREC 513 Idea Evaluation                    | 2  | 7  |
| 2024 | Fall        | AREC 511 Opportunities in Ag Value Chain    | 2* | 5  |
| 2024 | Summer      | AREC 586B New Venture Launch Practicum†     | 2  | 11 |
| 2024 | Summer      | AREC 520 Intel. Prop. in Food and Ag.       | 2  | 11 |
| 2024 | Spring AREC | AREC 586A New Venture Launch Practicum†     | 2  | 12 |
| 2024 | Spring      | AREC 518 Access to Capital in Ag. Sector    | 2* | 11 |
| 2024 | Spring      | AREC 478 Agricultural Policy                | 3  | 20 |
| 2024 |             | Total                                       | 22 |    |
| 2023 | Fall        | AREC 517 Entrepreneurial Identity & Teams † | 2  | 11 |
| 2023 | Fall        | AREC 513 Idea Evaluation†                   | 2  | 12 |
| 2022 | Fall        | AREC 511 Opportunities in Ag Value Chain    | 2* | 12 |
| 2023 | Summer      | AREC 520 Intel. Prop. in Food and Ag. †     | 2  | 3  |
| 2023 | Spring      | AREC 518 Access to Capital in Ag. Sector†   | 2* | 5  |
| 2023 | Spring      | AREC 478 Agricultural Policy                | 3  | 29 |
| 2023 |             | Total                                       | 13 |    |
| 2022 | Fall        | AREC 511 Opportunities in Ag Value Chain†   | 2* | 6  |
| 2022 | Spring      | AGRI/IE 270 World Interdependence           | 3  | 94 |
| 2022 | Spring      | AREC 478 Agricultural Policy                | 3  | 20 |
| 2022 |             | Total                                       | 8  |    |
| 2021 | Spring      | AGRI/IE 270 World Interdependence           | 3  | 56 |
| 2021 | Spring      | AREC 478 Agricultural Policy                | 3  | 36 |
| 2021 |             | Total                                       | 6  |    |
| 2020 | Spring      | AGRI/IE 270 World Interdependence           | 3  | 70 |
| 2020 | Spring      | AREC 478 Agricultural Policy                | 3  | 36 |
| 2020 | Spring      | AREC 505 Economics of Production            | 3  | 5  |
| 2020 |             | Total                                       | 9  |    |

† new courses developed for the first time for the Masters in Agribusiness and Food Innovation Management, taught in person at CSU Spur.

\* co-instructed course, shared teaching load

## STUDENT ADVISING/GRADUATE SUPERVISION

### GRADUATE STUDENTS:

#### Current PhD Advisees:

Sarah Blandon, Ph.D., Graduate Degree Program in Ecology (GDPE), with field in Agricultural and Resource Economics

**Current PhD Committee Memberships (excluding those chaired):**

Per Hulthén, Ph.D., Technology Management and Economics, with field in Entrepreneurship and Strategy, Chalmers University of Technology. Advisor: Mats Lundqvist.

**Graduate Degrees Completed Under Supervision:**

Fatemeh Lashgary, M.A.F.I.M., December 2025.

Elodie Kouadio, M.A.F.I.M., December 2025.

Babatunde Ojetunde, Ph.D., Horticulture, Colorado State University. Advisor: Jessica Davis. November 2025

Fred Jacobs, Masters in Agribusiness and Food Innovation Management (M.A.F.I.M.), December 2024.

Morgan Johnston, M.A.F.I.M., December 2024.

Rebecca Jurado, M.A.F.I.M., December 2024.

Madeline Livermore, M.A.F.I.M., December 2024.

Jordan Lobato, M.A.F.I.M., December 2024.

Ashley Murdock, M.A.F.I.M., December 2024.

Sarah Rhodes, M.A.F.I.M., December 2024.

Max Roepe, M.A.F.I.M., December 2024.

Juan Ignacio Sanchez Larrain, M.A.F.I.M., December 2024.

Sofia Valeriano Bubick, M.A.F.I.M., December 2024.

Matt West, M.A.F.I.M., December 2024.

Ulmashkhon Kalandarova, Ph.D., Economics, Colorado State University, Co-Advisors: Stephan Weiler and Anita Pena, December 2024.

Annabelle Berklund, Ph.D., Economics, Colorado State University. Advisor: Stephan Weiler. December 2024.

Aaisha Al Maamari, Ph.D., Agricultural and Resource Economics, May 2024.

Elbahlul Bdawi, Ph.D., Graduate Degree Program in Ecology, with field in Agricultural and Resource Economics, Colorado State University, May 2022.

Ghulam Samad, Ph.D., Graduate Degree Program in Ecology (GDPE), with field in Agricultural and Resource Economics, Colorado State University, August 2020.

Ahmed Alzahrani, M.S. (Plan B), Agricultural and Resource Economics, Colorado State University, August 2020.

Angelique Giraud, M.S. (Plan A), Agricultural and Resource Economics, Colorado State University, August 2019.

Bo Heiden, Ph.D., Technology Management and Economics, with field in Entrepreneurship and Strategy, Chalmers University of Technology, December 2017.

Leana Schwartz, M.Ag. (Plan B), Peace Corps Masters International (PCMI) program, with field in Agricultural and Resource Economics, August 2017.

Yoo Hwan Lee, Ph.D., Agricultural and Resource Economics, Colorado State University, August 2016.

Paouya Madani, M.S., Chalmers School of Entrepreneurship, Chalmers University of Technology, June 2016.

Joakim Olsson, M.S., Chalmers School of Entrepreneurship, Chalmers University of Technology, June 2016.

Robert Smolander, M.S., Chalmers School of Entrepreneurship, Chalmers University of Technology, June 2016.

Neama Lariel, Ph.D., Graduate Degree Program in Ecology, with field in Agricultural and Resource Economics, Colorado State University, August 2015.

Natalia Lukicheva, M.S., Agricultural and Resource Economics, Colorado State University, May 2015.

Jakrapun Suksawat, Ph.D., Graduate Degree Program in Ecology, with field in Agricultural and Resource Economics, Colorado State University, August 2015.

Chuba Suntharlingam, Ph.D., Graduate Degree Program in Ecology, with field in Agricultural and Resource Economics, Colorado State University, August 2014.

Kanokkon Pattarachon, M.S., Agricultural and Resource Economics, Colorado State University, August 2013.

## OTHER ACTIVITIES/ACCOMPLISHMENTS IN TEACHING/ADVISING

**Masters of Agribusiness and Food Innovation Management.** The eleven members of the first graduating cohort were granted their master's degrees in December 2024. Two members of the second graduating cohort were granted their master's degrees in December 2025. The causes of the downturn in enrollments for the second cohort are still not clear. But that factor, combined with new budgetary constraints under the new administration in 2025, led to a decision to put the MAFIM program on pause, and no new students were enrolled for the third cohort that would have started in August 2025.

Launched in August 2022, this program was designed for professionals from a variety of academic, business, and personal backgrounds, this intensive program develops the suite of creative and business skills needed to drive innovation and development of new enterprises in today's agricultural and food industries. This in-person degree program is offered at Colorado State University's new Spur campus, located at the National Western Center in Denver, as of Fall 2022. It is a 16-month full-time degree or 28 month part-time degree.

Through a combination of coursework, active networking, and practical experience, the program provides training and hands-on experience in developing and launching new products, technologies, and businesses. Students engage in coursework that immerses them in:

- The inner workings and latest developments of the agricultural and food system
- Essential business topics including accounting, finance, intellectual property, and regulatory compliance

Then, students gain hands-on, real-world experience through a unique, integrative, team-based, business development practicum project. Students choose one of two practicum tracks. In both tracks, student teams are paired with external partners who are active innovators in science or industry. And they are guided through the practicum by CSU faculty, advisors, and entrepreneur-in-residence mentors.

- Venture creation track—Students form a team of 3 or 4 members to help analyze a new business prospect or invention by their external partner, develop a business plan, seek to raise capital, and prepare to launch a startup to commercialize the idea.
- Corporate partner track— Students form a team of 3 or 4 members to engage with an existing company in something like a consulting role, to help analyze and develop an innovation within that company's current offerings or as a possible new line of business.

The practical, hands-on nature of this program offers opportunities for students to meet their career goals in a variety of ways. In some cases, students may be successful in launching a startup out of their practicum project and could take a financial stake and even a leadership role in their new company. In other cases, students and their corporate partners may find there is an excellent fit for further employment at the company after their practicum project is complete. In all cases, students launch from this program into the professional world with new relationships, skills, and knowledge that make them highly competitive innovators and entrepreneurs in today's rapidly evolving agriculture and food industries.

**USDA Faculty Exchange Program (FEP).** With PI Bill Spencer (CSU, College of Agricultural Sciences) and Co-PI John Olienyk (CSU, College of Business). I served for several years—from 2012 until the program at CSU ended in 2015—as a faculty mentor to program participants from the Ukraine, Kazakhstan, and other former Soviet republics. This international professional development program annually hosted 4 to 6 early-career faculty from post-Soviet republics for the Fall semester at CSU, to observe courses in agribusiness, agricultural economics, business management, finance, and marketing,

and to learn about the workings of the U.S. agricultural industry. Each Spring semester, a team of faculty members from CSU then travel to the participants' home countries and home institutions, in order to follow up with them, their supervisors, and senior administration, in order to help advance the participants' career development.

## **CV SECTION 4: OUTREACH/SERVICE**

### **COMMITTEES**

#### **University Committee, year(s)**

|             |                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 - 2025 | President's Spur Education Task Force, Office of the President and Provost, convened by James Pritchett and Marion Underwood, chaired by Jim Bradeen and Paula Mills |
| 2020 - 2023 | Committee on Innovation Strategies (CIS), Office of Vice-President of Research                                                                                       |
| 2016 - 2022 | Committee on Scholarship, Research and Graduate Education (CoSRGE), Faculty Council and Graduate School                                                              |
| 2012 - 2014 | Value Chain of Colorado Agriculture research and engagement team, Agriculture and Food key industry task force for the Colorado Blue Print, CSU Office of Engagement |

#### **College Committees, year(s)**

|             |                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------|
| 2016 - 2022 | CSU Spur/National Western Center agricultural innovation initiatives                                         |
| 2014 - 2017 | CSU Agricultural Innovation Summit steering committee                                                        |
| 2012 - 2015 | External Advisory Board Member, Global Social and Sustainable Enterprise (GSSE) program, College of Business |
| 2010 - 2011 | Search committee, Bioenergy cluster hire, chaired by Lee Sommers                                             |

#### **Department Committees, year(s)**

|                |                                                            |
|----------------|------------------------------------------------------------|
| 2025 – present | REACH OUT Committee                                        |
| 2019 - 2024    | Justice, Equity, Diversity, and Inclusion (JEDI) Committee |
| 2015 - 2016    | Ag Policy PhD Field Exam Committee                         |
| 2009 - 2014    | Microeconomics PhD Core Exam Committee                     |
| 2013 - 2014    | Graduate Admissions Committee                              |
| 2012 - 2013    | Graduate Curriculum Review Committee                       |

### **PROFESSIONAL AFFILIATIONS AND ACTIVITIES**

Clearly differentiate editorial positions (e.g. journal editor/associate editor) from manuscript and grant refereeing.

### Memberships in professional societies

|                |                                                      |
|----------------|------------------------------------------------------|
| 1999 - present | American Applied Economics Association (AAEA)        |
| 1999 - 2020    | American Economics Association (AEA)                 |
| 2000 - 2011    | Association of University Technology Managers (AUTM) |
| 2000 - 2003    | Licensing Executives' Society (LES)                  |

### Service to professional societies

|             |                                                                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019 – 2025 | Chair, of Agricultural Experiment Station multistate research project NC1034, “ <i>Impact Analyses and Decision Strategies for Agricultural Research</i> ” |
| 2018 - 2020 | Selected Presentations Committee, AAEA                                                                                                                     |
| 2017        | Advisory Committee, Convening on Innovation Adoption Across the Food Chain, FFAR                                                                           |
| 2013 - 2015 | Outstanding MS Awards Committee, AAEA                                                                                                                      |

### Grant review panels

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| 2022 – 2025 | CSU Strata, internal reviewer for Colorado OEDIT Advanced Industries Proof of Concept grants program          |
| 2018 - 2019 | Genome Canada, 2019 Large-Scale, Agricultural Research Project Competition, Pre-Application Review Committee  |
| 2014 - 2015 | Genome Canada, 2014 Large-Scale, Agricultural Research Project Competition, Full Application Review Committee |

### Grant proposal reviews

|      |                             |
|------|-----------------------------|
| 2017 | National Science Foundation |
|------|-----------------------------|

### Journal refereeing

|      |                                                        |
|------|--------------------------------------------------------|
| 2025 | <i>Food Policy</i>                                     |
|      | <i>Applied Economic Perspectives and Policy (AEPP)</i> |
| 2024 | <i>Applied Economic Perspectives and Policy (AEPP)</i> |
|      | <i>Agribusiness</i>                                    |
|      | <i>Food Policy</i>                                     |
| 2023 | <i>Food Policy</i>                                     |
|      | <i>Applied Economic Perspectives and Policy (AEPP)</i> |
|      | <i>Yale Journal of Health Policy, Law, and Ethics</i>  |

|      |                                                                 |
|------|-----------------------------------------------------------------|
| 2022 | <i>Handbook of Agricultural Economics</i>                       |
| 2021 | <i>American Journal of Agricultural Economics (AJAE)</i>        |
|      | <i>Sustainability</i>                                           |
|      | <i>Journal of World Intellectual Property</i>                   |
|      | <i>Agronomy Journal</i>                                         |
|      | <i>Yale Journal of Health Policy, Law, and Ethics</i>           |
| 2018 | <i>Journal of Entrepreneurship and Environmental Management</i> |
|      | USDA Economic Research Service report                           |
|      | <i>Agricultural Economics</i>                                   |
|      | <i>Australian Journal of Grape and Wine Research</i>            |
|      | <i>Journal of Agricultural and Resource Economics (JARE)</i>    |
| 2017 | USDA Economic Research Service report                           |